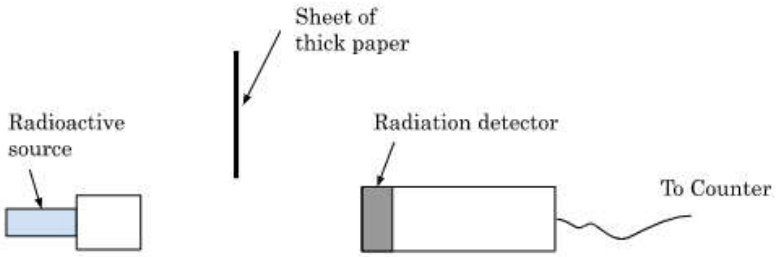
Question 1 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

A student uses the arrangement shown below to ascertain the type of radiation emitted by a given sample of radioactive material.



When the sheet of paper is in the position shown, a count rate of 720 Bq is recorded. When she lowers the thick sheet of paper so as to obstruct the direct path between the source and the detector, the count rate drops to 18 Bq. Which of the following is the most likely explanation for the data collected?

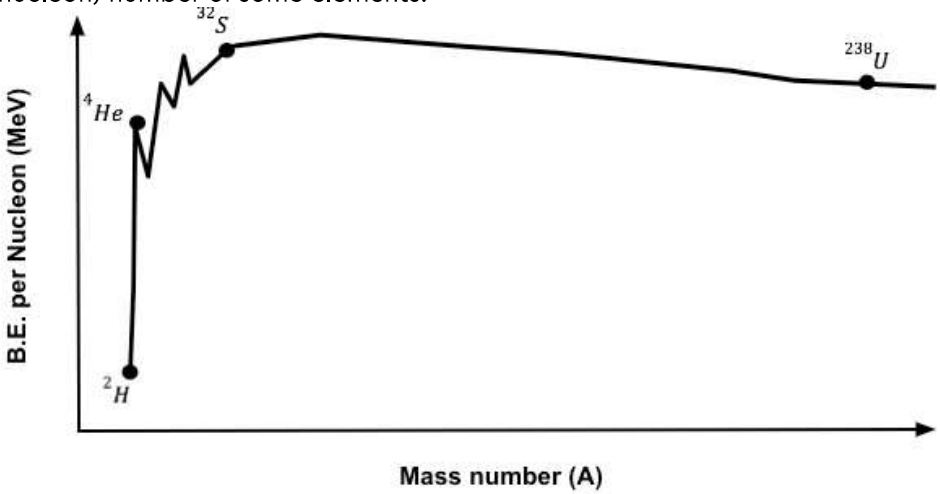
- A. Only alpha particles are emitted. The remaining activity is from background radiation.
- B. An equal number of alpha and beta particles are emitted.
- C. Only beta particles are emitted.
- D. There is not enough data to draw a reasonable conclusion.

Question 2 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

The diagram below shows the graph of binding energy per nucleon vs. the mass (nucleon) number of some elements.



Which is the most stable nucleus?

- A. $^{238}\mathrm{U}$
- B. $^{2}\mathrm{H}$
- C. $^{4}\mathrm{He}$
- D. $^{32}\mathrm{S}$

Question 3 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

Which statement describes a possible nuclear fusion reaction?

- A. Hydrogen nuclei join together to form a helium nucleus.
- B. A helium nucleus joins with a hydrogen nucleus to form an alpha particle.
- C. Uranium nuclei split and produce high energy neutrons causing a chain reaction.
- D. Helium nuclei join to form hydrogen nuclei.

Question 4 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

A nuclear reactor converts nuclear energy into electrical energy. Which of the following reactions best describes the primary nuclear reaction in a nuclear reactor?

- A. $^{234}_{90}\mathrm{Th} \rightarrow {}^{0}_{-1}\beta + {}^{234}_{91}\mathrm{Pa} + {}^{0}_{0}\overline{\nu}$
- B. $^{232}_{92}\mathrm{U} \rightarrow {}^{228}_{90}\mathrm{Th} + {}^{4}_{2}\alpha$
- C. $^{235}_{92}\mathrm{U} + {}^{1}_{0}n \rightarrow {}^{90}_{36}\mathrm{Kr} + {}^{143}_{56}\mathrm{Ba} + 3\,{}^{1}_{0}n$
- D. $^{14}_{6}\mathrm{C} \rightarrow {}^{7}_{3}\mathrm{Li} + {}^{7}_{3}\mathrm{Li}$

Question 5 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

A single ^{37}Cl atom has a mass of 36.966 u. What is the mass of 0.5 moles of ^{37}Cl ?

- A. 18.483 grams
- B. 18.483 kilograms
- C. 36.966 grams
- D. 36.966 kilograms

Question 6 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

Which of the following correctly lists the fundamental forces in decreasing order of strength?

- A. Weak nuclear, strong nuclear, electromagnetic, gravitational
- B. Strong nuclear, electromagnetic, weak nuclear, gravitational
- C. Gravitational, electromagnetic, weak nuclear, strong nuclear
- D. Electromagnetic, gravitational, strong nuclear, weak nuclear

Question 7 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

Which of the following correctly represents the hadron type and quark combination for the particle listed?

	Particle	Hadron Type	Quark composition
A.	neutron	baryon	u u d
B.	neutron	meson	u d d

	Particle	Hadron Type	Quark composition
C.	proton	baryon	u u d
D.	positive pion	meson	u d

Question 8 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

Which of the following correctly states the conclusion drawn from the Rutherford-Geiger-Marsden experiment?

- A. The nucleus of an atom contains protons and neutrons.
- B. Protons and electrons have equal but opposite charges.
- C. The nucleus contains nearly the entire mass of the atom.
- D. Gold atoms absorb alpha particles.

Question 9 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

An electron in a hydrogen atom is excited from the ground state of $n = 1$ to the $n = 4$ level. What is the total number of frequencies that can be observed from the emission spectrum as the electron returns to ground state?

- A. 12
- B. 6
- C. 4
- D. 3

Question 10 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

A nuclide of a radioactive element Q can be represented as ${}^{234}_{90}\mathrm{Q}$. The nuclide undergoes two negative beta decays and one alpha decay. Which of the following configurations correctly represents the daughter nuclide R?

- A. ${}^{234}_{92}\mathrm{R}$
- B. ${}^{232}_{92}\mathrm{R}$
- C. ${}^{230}_{90}\mathrm{R}$
- D. ${}^{230}_{92}\mathrm{R}$

Question 11 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

Consider the following statements about nuclear binding energy:
The nuclear binding energy is:

- I. proportional to the mass defect from nucleus formation.
- II. the energy released when a nucleus is formed from its nucleons.
- III. the energy required to remove a proton from the nucleus.

Which of the statements are correct?

- A. I only
- B. I and II only
- C. II only

- D. I, II and III

Question 12 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

The binding energy of ^4_2He is 28 MeV. What is its binding energy per nucleon?

- A. 45×10^{-13} J
- B. 22×10^{-13} J
- C. 11×10^{-13} J
- D. 7.5×10^{-13} J

Question 13 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

In the absence of the strong nuclear force, a large nucleus would be even more unstable than a small nucleus. Which one of the following is the best explanation?

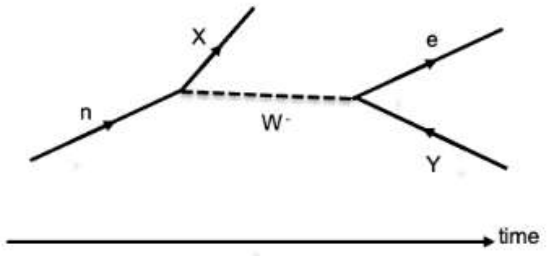
- A. A nucleus with fewer protons will more likely stay together due to the gravitational attraction between the nucleons.
- B. The more protons in the nucleus, the more repulsive force there would be to make it unstable.
- C. A nucleus with fewer neutrons will more likely stay together due to the gravitational attraction between the nucleons.
- D. The more neutrons there are in the nucleus, the larger the separation of the protons which weakens the electrostatic repulsion between the protons.

Question 14 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

The Feynman diagram shows the decay of a neutron.
What are the particles labeled X and Y?



	X	Y
A.	proton	neutrino
B.	neutron	neutrino
C.	proton	antineutrino
D.	neutron	antineutrino

Question 15 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

Which of the following is not true about gluons?

- A. Gluons are massive particles.
- B. Gluons travel at the speed of light.

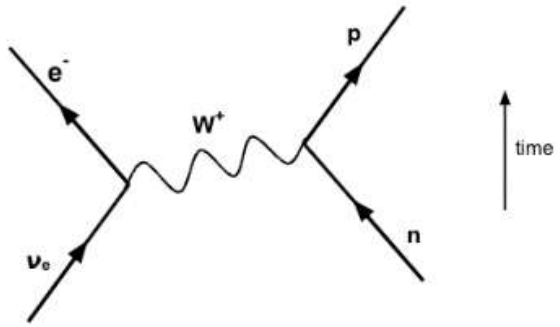
- C. Gluons do not carry an electrical charge.
- D. Gluons mediate the strong force between quarks.

Question 16 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

The diagram below shows the Feynman diagram for an interaction between two particles.



Which of the equations below correctly represents the reaction?

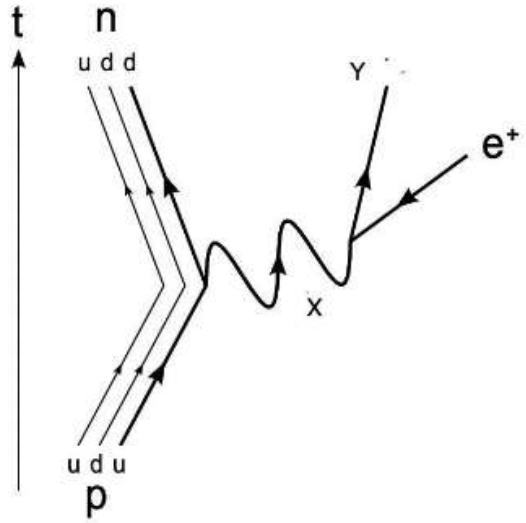
- A. $n + v_e \rightarrow p + e^-$
- B. $p + e^- \rightarrow n + v_e$
- C. $v_e + e^- \rightarrow n + p$
- D. $(v_e + e^-) - (n + p) \rightarrow W^+$

Question 17 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

The picture below shows the Feynman diagram of a β^+ decay.



[Adapted from the original image at Wikicommons]

Identify the particles X and Y.

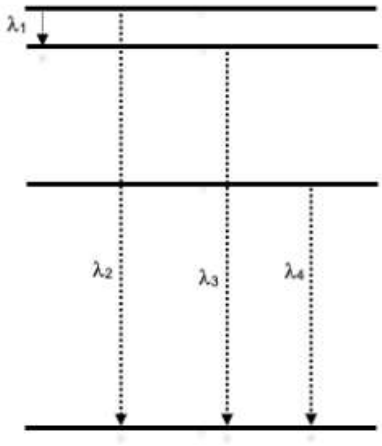
	X	Y
A.	W^+ boson	electron neutrino
B.	W^+ boson	electron antineutrino
C.	Z^0 boson	electron neutrino
D.	Z^0 boson	electron antineutrino

Question 18 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

The diagram shows atomic energy levels of an element, and wavelengths are labeled corresponding to emission for electron transition between certain energy levels.



Which of the following shows these wavelengths in decreasing order?

- A. $\lambda_1, \lambda_2, \lambda_3, \lambda_4$
- B. $\lambda_4, \lambda_3, \lambda_2, \lambda_1$
- C. $\lambda_1, \lambda_4, \lambda_3, \lambda_2$
- D. $\lambda_2, \lambda_3, \lambda_4, \lambda_1$

Question 19 Calculator Allowed

Markscheme

Difficulty: **Easy**

[Maximum marks: 6]

A thorium-234 ($\mathrm{Th\text{-}234}$) nucleus undergoes beta decay and transforms into a protactinium-234 nucleus ($\mathrm{Pa\text{-}234}$). The binding energies per nucleon for Thorium-234 and protactinium-234 are $7.84\ MeV$ and $7.92\ MeV$ respectively. An electron has a rest mass energy of $0.020\ MeV$.

The reaction can be represented as follows:

$^{234}_{90}\mathrm{Th} \rightarrow\ ^{234}_{91}\mathrm{Pa} + 10e^- + \overline{\nu_e}$

1. Explain what is meant by binding energy per nucleon. [2]
2. Calculate, in joules, the total energy released in the decay. [3]
3. State the name of the exchange particle in the reaction. [1]

Question 20 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

Which of the following radioactive decays can cause the production of a daughter nucleus to be a different element than the parent nucleus?

- I. α decay
  II. β decay
  III. γ decay

- A. I only

- B. II only
- C. I and II only
- D. I, II, and III

Question 21 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

A 1.20 kg sample of a radioactive isotope decaying via beta decay with a half-life of 30 minutes is observed in a closed container in a laboratory. The decay results in a daughter nucleus that is also a solid. What is the approximate total mass measured after 90 minutes?

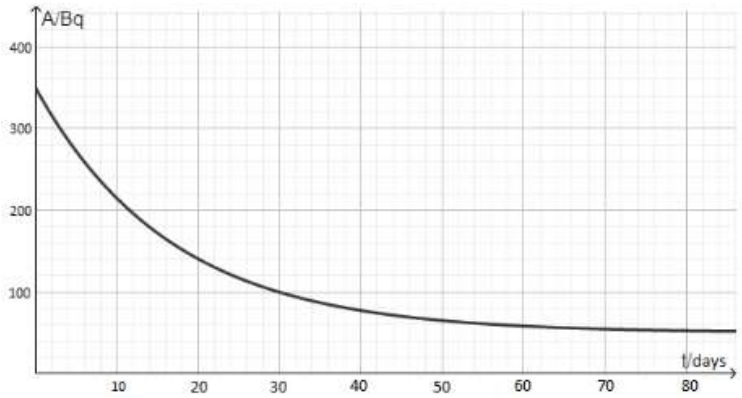
- A. 0.00 kg
- B. 0.15 kg
- C. 0.60 kg
- D. 1.20 kg

Question 22 Calculator Allowed

Markscheme

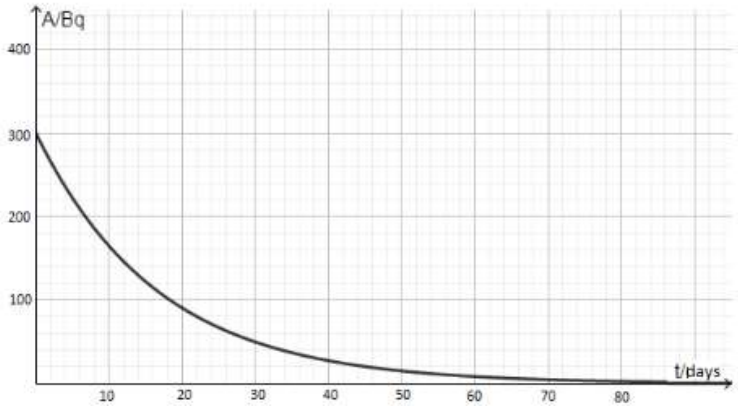
Difficulty: **Easy**

[Maximum mark: 6]
The graph shows the variation of the measured activity of an isotope sample with time.



[Source: Created with GeoGebra, - https://www.geogebra.org/graphing?lang=en]

1.
State what is meant by activity. [1]
2.
Explain why the activity graph doesn't approach zero as time increases. [2]
3.
The activity graph is corrected so as to reach zero as time increases as shown below.



[Source: Created with GeoGebra, - https://www.geogebra.org/graphing?lang=en]

1.

Determine the half-life of the sample. [1]

2.

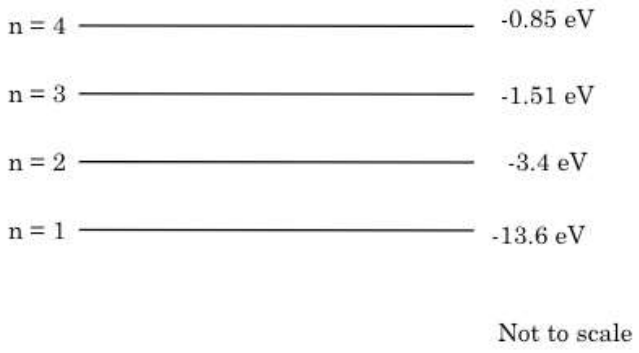
The initial mass of this sample was 248 g. Determine the time that would elapse before 31 g of the sample remained. [2]

Question 23 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

The electron in the ground state of a hydrogen atom absorbs a certain amount of energy. The diagram shows some of the energy levels of the hydrogen atom. At which state would the electron be if it absorbs 12.09 eV of energy?



- A. At ground state.
- B. First excited state.
- C. Second excited state.
- D. Third excited state.

Question 24 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

Which of the following lists three fundamental forces in decreasing order of range?

- A. weak, strong, electromagnetic
- B. strong, weak, gravitational
- C. electromagnetic, strong, weak
- D. gravitational, weak, strong

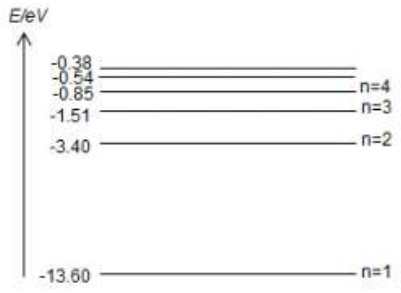
Question 25 Calculator Allowed

Markscheme

Difficulty: **Medium**

[Maximum mark: 10]

The diagram shows six energy levels of hydrogen.



1.

Describe how an emission spectrum is an evidence of discrete energy levels in an atom. [2]

2.

1. On the energy level diagram, draw an arrow to represent the transition that emits the shortest wavelength. [2]
2. Calculate the shortest emitted wavelength. [3]

3. Emitted red light has a frequency estimation of 460 THz. Determine which energy level transition emits this red light. [3]

Question 26 Calculator Not Allowed

Markscheme

Difficulty: **Medium**

The binding energy per nucleon of a ${}^{A}_{Z}\mathrm{X}$ nucleus is E. What is the mass defect for the formation of X from its constituent nucleons?

- A. $\frac{AE}{c^2}$
- B. $\frac{ZE}{c^2}$
- C. $\frac{E}{Ac^2}$
- D. $\frac{E}{Zc^2}$

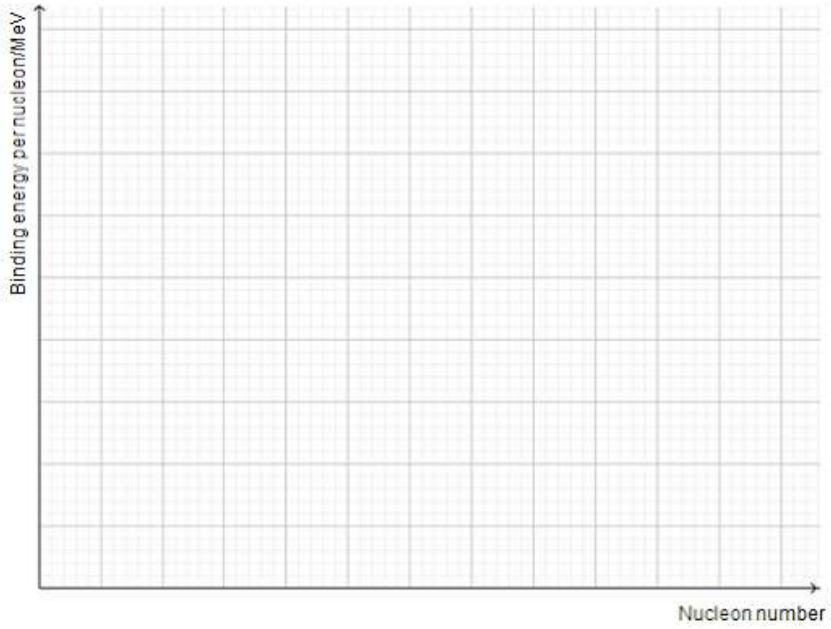
Question 27 Calculator Allowed

Markscheme

Difficulty: **Medium**

[Maximum Mark: 10]

1. Uranium-235 (${}^{235}_{92}U$) has a nuclear mass of 235.044 u.
   1. State what is meant by mass defect. [1]
   2. Calculate the mass defect of uranium-235. [2]
2.
   1. Calculate the binding energy per nucleon of uranium-235. [2]
   2. On the axes given, sketch the variation of the binding energy per nucleon with the nucleon number. Label the position of uranium-235. There is no need to add numbers to the axes. [3]



[Source: Created with GeoGebra (https://www.geogebra.org/graphing?lang=en)]

3.
Uranium-235 ($^{235}_{92}U$) decays into thorium-231 ($^{231}_{90}Th$).

- Determine which nuclide has a higher binding energy per nucleon. Use the graph to justify your answer. [2]

Question 28 Calculator Not Allowed

Markscheme

Difficulty: **Medium**

A nucleus $^{A}\mathrm{X}$ has a mass defect of N unified atomic mass units, Nu. What is the correct expression for its binding energy per nucleon?

- A. $931.5\frac{N}{A}\ MeV$
- B. $931.5\frac{N}{A}\ MeV\ c^{-2}$
- C. $931.5\frac{A}{N}\ MeV$
- D. $931.5\frac{N}{A}\ MeV\ c^{-2}$

Question 29 Calculator Allowed

Markscheme

Difficulty: **Medium**

1.
The following reaction takes place in a nuclear reactor:

$$^{235}_{92}\mathrm{U} + {}^{1}_{0}n \rightarrow {}^{144}_{56}\mathrm{Ba} + {}^{89}_{36}\mathrm{Kr} + 3{}^{1}_{0}n$$

1.
Differentiate between fission and fusion reactions. [3]

2.
Determine whether this reaction is a nuclear fission or nuclear fusion. [1]

2.
The following data is given for this reaction:

The binding energy per nucleon of $^{235}\mathrm{U}$ = 7.59 MeV

The binding energy per nucleon of $^{144}\mathrm{Ba}$ = 8.27 MeV

The binding energy per nucleon of $^{89}\mathrm{Kr}$ = 8.62 MeV

Calculate the energy released from this reaction. [3]

Question 30 Calculator Not Allowed

Markscheme

Difficulty: **Medium**

Which one of the following is the best explanation for why isolated quarks are not observed?

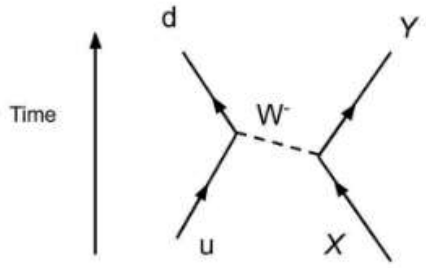
- A. Quarks are very close to each other, and gravitational forces bind several of them strongly together.
- B. Quarks are bound by a very strong force, so the energy required to pull them apart is enough to create new pairs of quarks.
- C. Due to strong forces quarks combine into larger particles called leptons.
- D. Quarks are extremely tiny particles and do not interact electromagnetically, so cannot be seen.

Question 31 Calculator Not Allowed

Markscheme

Difficulty: **Medium**

The Feynman diagram shows a particle interaction between an up quark and a lepton.



Which correctly labels particles X and Y?

	X	Y
A.	electron	electron neutrino
B.	electron	electron anti-neutrino
C.	positron	electron neutrino
D.	positron	electron anti-neutrino

Question 32 Calculator Not Allowed

Markscheme

Difficulty: **Medium**

An unstable nucleus decays. Which of the following could be involved in the interaction in the nucleus?

- I. Weak nuclear force
  II. Strong nuclear force
  III. Electromagnetic force

- A. I only
- B. II only
- C. I and II only
- D. I, II, and III

Question 33 Calculator Allowed

Markscheme

[Maximum mark: 10] Difficulty: **Medium**

Particle X^- is a meson with a strangeness of -1 and a charge of -1. It decays as shown:

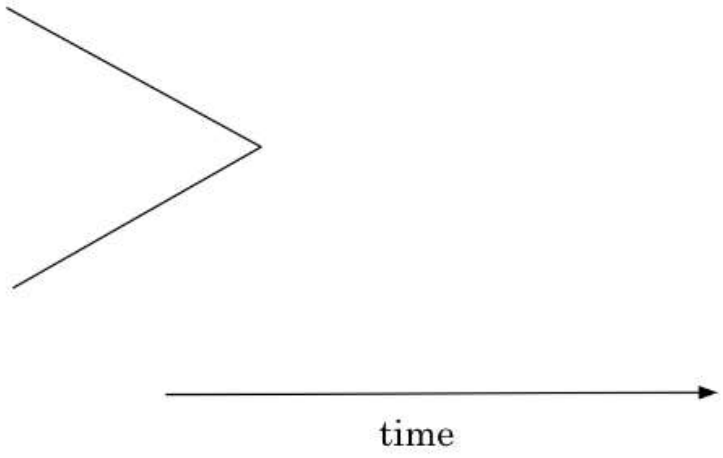
$X^- \rightarrow \mu^- + \overline{\nu_\mu}$

1.
Distinguish between baryons and mesons. [2]

2.
State the quark structure of X^-. Justify your answer. [3]

3.
Charge, energy and momentum are conserved in this reaction. State another quantity that is conserved, and a quantity that is not conserved. [2]

4.
Complete the Feynman diagram for this decay. The first vertex has been partially constructed. [3]



Question 34 Calculator Allowed

Markscheme

Difficulty: **Medium**

[Maximum mark: 10]

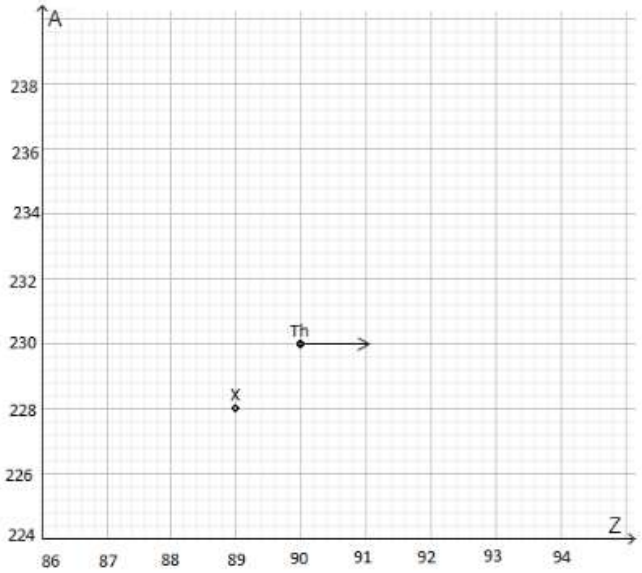
The plutonium isotope $^{238}_{94}\mathrm{Pu}$ undergoes alpha decay.

$^{238}_{\square}Pu \rightarrow ^{\square}_{92}U + ^{\square}_{\square}\square$

1.
Define isotope. [2]

2.
Rewrite this decay equation, filling in any missing information. [3]

3.
Plutonium undergoes a decay series. The graph below shows a part of this decay series.



[Source: Created with GeoGebra, - https://www.geogebra.org/graphing?lang=en]

1.
One type of decay cannot be represented on this diagram. Outline this type of decay and justify your answer. [2]

2.
State the type of decay of $^{230}_{90}\mathrm{Th}$ that is represented on the diagram above. Justify your answer. [2]

3.
Isotope X undergoes an alpha decay followed by a beta minus decay to produce isotope Y. Label the position of isotope Y on the diagram. [1]

4.
Compare the penetrating powers of alpha, beta and gamma radiation. [1]

Question 35 Calculator Not Allowed

Markscheme

Difficulty: **Medium**

Which of the following correctly shows the change in total binding energy per nucleon in nuclear fission and nuclear fusion?

	Nuclear fission	Nuclear fusion
A.	increases	increases
B.	decreases	increases
C.	decreases	decreases
D.	increases	decreases

Question 36 Calculator Allowed

Markscheme

Difficulty: **Medium**

[Maximum marks: 9]

1.
A thorium-234 ($\mathrm{Th\text{-}234}$) nucleus undergoes beta decay and transforms into a protactinium-234 nucleus ($\mathrm{Pa\text{-}234}$). The binding energies per nucleon for Thorium-234 and protactinium-234 are $7.84\ MeV$ and $7.92\ MeV$ respectively. An electron has a rest mass energy of $0.020\ MeV$.
The reaction can be represented as follows:
$^{234}_{90}\mathrm{Th} \rightarrow\ ^{234}_{91}\mathrm{Pa} + 10e^{-} + \overline{\nu_e}$

1.
Explain what is meant by binding energy per nucleon. [2]

2.
Calculate, in joules, the total energy released in the decay. [3]

3.
State the name of the exchange particle in the reaction. [1]

2.
A cubic container is filled with pure $\mathrm{thorium\text{-}234}$. The container does not contain any other elements. After 62 days, the ratio of the number of nuclei of decay product to $\mathrm{thorium\text{-}234}$ nuclei is 5.
Calculate, in days, the half-life of the $\mathrm{thorium\text{-}234}$ nuclei. [3]

Question 37 Calculator Not Allowed

Markscheme

Difficulty: **Medium**

The following conservation laws apply to particle interactions.

- I. Law of conservation of charge.
  II. Law of conservation of baryon number.
  III. Law of conservation of lepton number.

Given the impossible reaction:

$$P^+ + n^0 \rightarrow P^+ + \pi^0$$

Which of the conservation laws are violated in the reaction?

- A. I only
- B. II only
- C. III only
- D. I, II, and III

Question 38 Calculator Allowed

Markscheme

Difficulty: **Medium**

[Maximum mark: 7]

1.
A kaon that has a quark structure of $u\bar{s}$ decays into an antimuon and muon neutrino.

$$K^+ \rightarrow \mu^+ + \nu_\mu$$

1.
Explain how it can be deduced that this decay happens through weak interaction. [2]

2.
Using the quark structure for the kaon, draw a Feynman diagram for this decay. [2]



2.
Explain, by reference to properties of nuclear and electrostatic forces, why small stable nuclei have equal numbers of protons and neutrons but in large nuclei there is an imbalance in the number of protons and neutrons. [3]

Question 39 Calculator Not Allowed

Markscheme

Difficulty: **Medium**

An element X is bombarded with alpha particles and a new element Y is formed with a neutron emitted. The following equation can be written for the nuclear reaction.

$$^{4}_{2}\mathrm{He} + {}^{\square}_{A}\mathrm{X} \rightarrow {}^{\square}_{B}\mathrm{Y} + \mathrm{n}$$

The atomic numbers of X and Y are A and B respectively. Which of the following is correct for A and B?

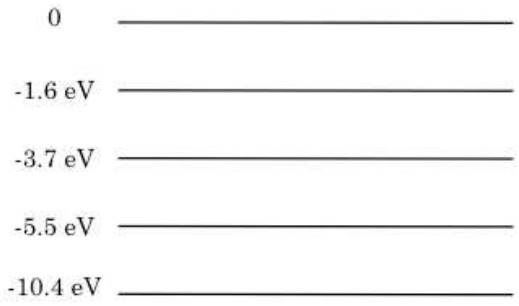
- A. $A - 1 = B$
- B. $A + 1 = B$
- C. $A - 2 = B$
- D. $A + 2 = B$

Question 40 Calculator Not Allowed

Markscheme

Difficulty: **Medium**

Some of the energy levels of electrons of a mercury atom are as shown.



Not to scale

Electrons and photons are incident on a mercury vapour causing energy level transitions. Emitted photons due to the transition of electrons between energy levels from the vapour have an energy of 4.9 eV.
Which of the following could be incident into mercury vapour?

- I. An electron with an energy of 5.1 eV
  II. A photon with an energy of 4.9 eV
  III. A photon with an energy of 5.1 eV

- A. I only
- B. II only
- C. I and II only
- D. II and III only

Question 41 Calculator Allowed

Markscheme

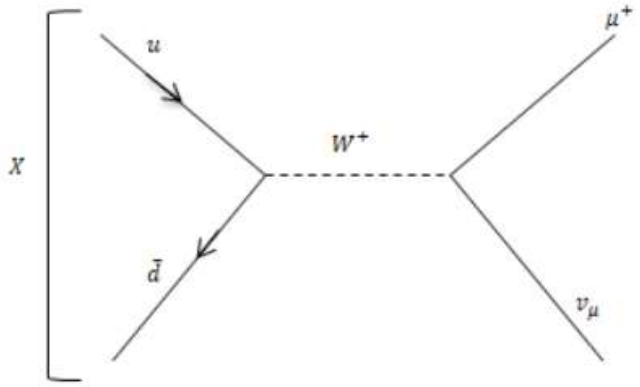
Difficulty: **Medium**

[Maximum mark: 10]
The equation represents the collision of a proton with a K^0 particle

$$K^0 + p \rightarrow n + \pi^+$$

The neutral kaon K^0 has a strangeness of $+1$, and its quark composition is $\bar{s}d$.

1.
Determine the quark structure of the π^+, given the fact that the strangeness is not conserved. Show your reasoning [3]
2.
1. The Feynman diagram represents the decay of particle X



Add the missing arrows and state the nature of particle X. [1][3]
1. State the equation represented by this Feynman diagram.

1.
The following equation is an impossible decay of K^0 kaon.
$K^0 \rightarrow \mu^+ + \overline{\nu_\mu}$
State one reason that makes this decay not possible. [1]
2.
Differentiate between hadrons and leptons. [2]

Question 42 Calculator Not Allowed

Markscheme

Difficulty: **Hard**

In a nuclear fission reaction, a nucleus of element X absorbs a neutron and decays into a nucleus of element Y and a nucleus of element Z.

$$X + n \rightarrow Y + Z + 2n$$

Which of the following statements are true about the binding energies are likely true:

- I. Binding energy per nucleon of Y is greater than the binding energy per nucleon of X.
- II. Total binding energy of Y and Z is less than the total binding energy of X.
- III. Total binding energy of Y and Z is greater than the total binding energy of X.
- IV. Binding energy per nucleon of Y is less than the binding energy per nucleon of X.

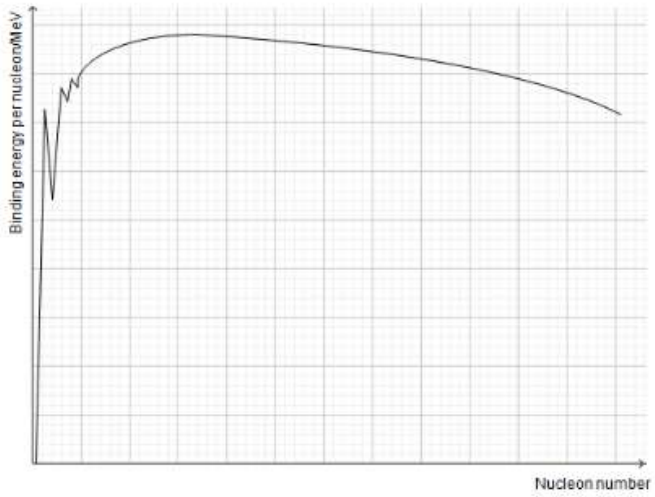
- A. Only I
- B. I and III
- C. Only IV
- D. II and IV

Question 43 Calculator Allowed

Markscheme

Difficulty: **Hard**

[Maximum mark: 10]
The graph shows the variation of the binding energy per nucleon with the mass number A.



[Source: Created with GeoGebra - https://www.geogebra.org/graphing?lang=en]

1.
State what is meant by binding energy. [1]
2.
Nickel-62 is energetically stable; it doesn't release energy by fission or fusion.
Label the position of Nickel-62 on the graph above. [1]
3.
The following reaction represents a nuclear fusion reaction:

$$ {}^{2}_{1}H + {}^{2}_{1}H \rightarrow {}^{3}_{2}He + {}^{A}_{Z}X $$

1.
State the nature of X along with its values for A and Z. [2]
2.
The following data is given about the reaction:

The nuclear mass of H-2 = 2.014 u

The nuclear mass of He-3 = 3.016 u

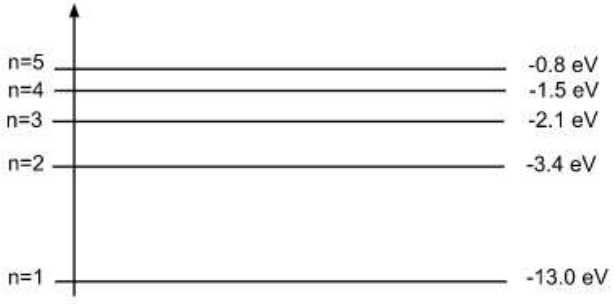
Calculate the energy released from the reaction. [3]
3.
All the energy released from the reaction goes into kinetic energy.
Assume that both of the ${}^{2}_{1}H$ particles are initially at rest. Determine the speed of X. [3]

Question 44 Calculator Not Allowed

Markscheme

Difficulty: **Hard**

The diagram gives energy levels of a single-electron atom.



How many different wavelengths can be detected in the emission spectrum of the atom?

- A. 4
- B. 5
- C. 9
- D. 10